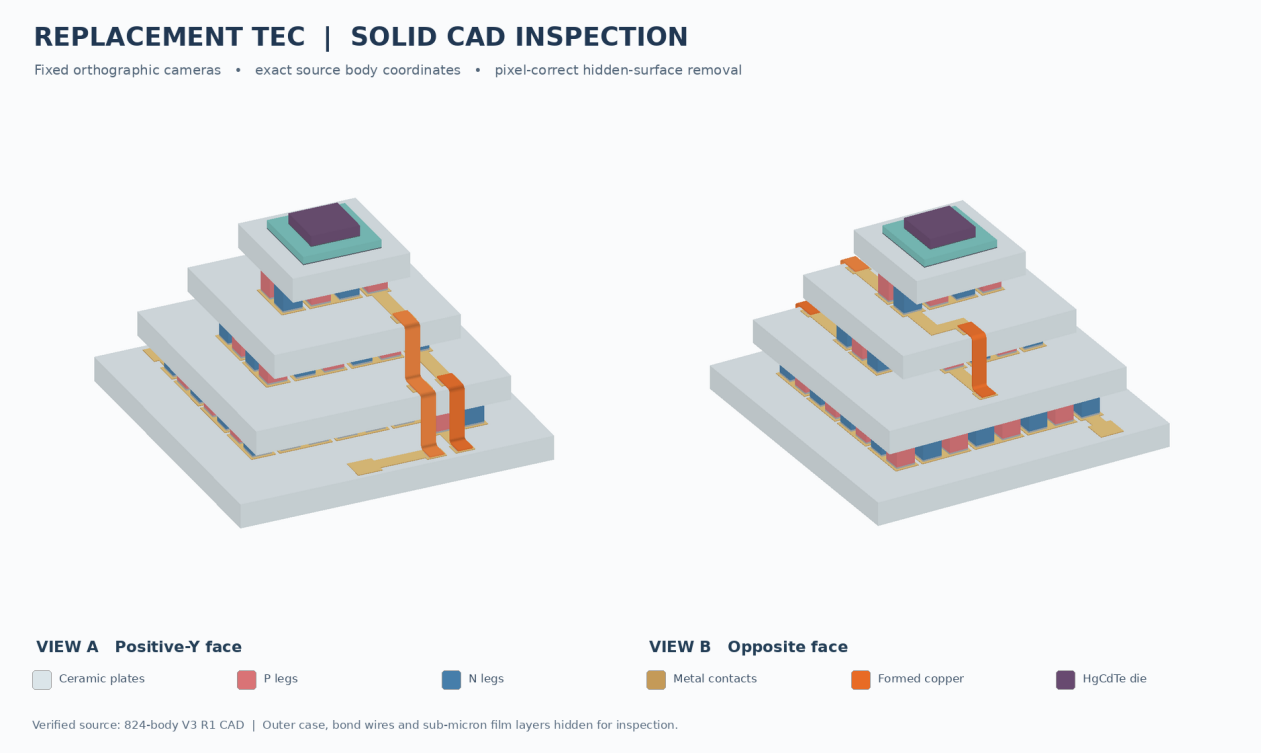
Thermoelectric Cooling of an HgCdTe Detector

Device Design, Heat Flow, and Electrical Performance

Three-stage cooler in a nine-pin package | 8 November 2026

PHYSICAL SCOPE. A three-stage cooler with a HgCdTe detector is evaluated using literature-informed thermoelectric properties, explicitly chosen package and joint assumptions, and calculated heat and current flow. The results describe a synthetic reference device, not a measured commercial product. Interstage foil heat leakage is now included through a conditional, source-CAD-based electrothermal multiport. True three-dimensional metal current spreading, deposited-film heat flow and measured joint impedances remain unresolved.



Proposed three-stage cooler: alumina substrates, thermoelectric legs, patterned electrode surfaces, and supported copper foil transitions. The package cover is removed to expose the heat-flow architecture. Image derived from the engineering geometry; materials and joints are design assumptions.

Evidence standard. Calculated performance, material-property assumptions, source dimensions, and unresolved physical losses are separated throughout. A low residual demonstrates energy accounting, not measured detector performance.

Engineering summary

Question. How cold can a compact, three-stage thermoelectric cooler hold an HgCdTe photoconductor when heat must leave through a real package, and what thermal or electrical losses most threaten its performance?

A synthetic cooler is examined with four alumina substrates, 44 P/N thermoelectric couples divided among three stages (29 hot-side, 11 middle, 4 cold-side), soldered contacts, patterned metallization, supported formed copper-foil electrical transitions, and a HgCdTe die mounted at the cold end. The device occupies a proposed nine-pin metal package with a window. The interior is *physically designed*, but its material grades and joint specifications are not measurements of a commercial cooler.

Nominal operating condition: 0.400 A applied TEC current; 300 K ambient thermal reservoir; 2.00 mW imposed detector heat; 100 Pa krypton in the enclosure; literature-derived P/N transport curves and selected contact/heat-sink parameters. The calculated operating point is

$$\boxed{T_{\text{detector}} = 218.833\ \text{K}} \qquad \boxed{V_{\text{TEC}} = 1.489\ \text{V}} \qquad \boxed{P_{\text{electrical}} = 0.596\ \text{W}}.$$

This corresponds to an *approximately 81.17 K detector-to-reservoir temperature difference*, not a validated maximum TEC temperature drop or measured commercial specification. The preceding V3.3 simulation, which omitted the formed-copper connections, gave 210.625 K; a conservative copper multiport raises the calculated detector temperature by **8.208 K** under identical nominal settings. The original V3.3 cavity-off/on comparison remains useful but is a separate earlier model-generation experiment.

Most important physical qualification: The newer electrical design removes the long unsupported copper-bar arrangement in favor of short formed foil links and a three-foot return, with defined ceramic landing pads. The current thermal calculation includes individually modeled P/N legs, 3D heat spreading through the four ceramics, solder/contact *electrical* losses, and effective package heat exchange. V3.4 now incorporates the source-defined copper bridges and their three-foot return as an energy-conserving, finite-contact *reduced* electrical/thermal network coupled to each genuine source ceramic FEM pad. Its corrected temperature is 218.833 K, while true 3D metal current crowding, solder thermopower and measured material/contact properties remain unqualified. Neither the former 210 K result nor the new 219 K result constitutes hardware validation.

Engineering use: The calculation identifies where energy is pumped, dissipated, parasitically imported or rejected, including about 108.5 mW removed from the hotter substrate through Cu connections, of which about 87.0 mW enters the upper substrate; it predicts temperature, voltage and stage-by-stage heat loads under declared assumptions; and it provides a reusable starting point for evaluating metallurgy, foil geometry, contact thermal resistance, gas pressure and heat-sink design. Numerical mesh details and document-history evidence appear *after* the physical assessment.

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1 What the copper interconnects do to cooling performance

A thermoelectric cooler transfers heat toward its hot side while electrical interconnects must also cross between ceramic stages. The three short formed copper-foil connections in the current cooler provide a real electrical path, but they simultaneously create passive conductive paths that partially undo the cooling. This effect is absent from the original V3.3 prediction. A new V3.4 calculation adds the actual two-foot and three-foot link topology with physically separate ceramic-pad thermal terminals.

At the *same* 0.400 A input, 300 K heat reservoir, 2.00 mW detector heat and 100 Pa krypton setting:

Calculated quantity	V3.3	V3.4 with copper	Change
Detector temperature (K)	210.625214	218.832895	+8.207681
Terminal voltage (V)	1.522462	1.489440	-0.033023
Electrical input (W)	0.608985	0.595776	-0.013209
Sink heat rejection (W)	0.594044	0.579830	-0.014214

Although the predicted electrical power falls, the detector is **8.208 K warmer**. The important loss mechanism is *parasitic stage-to-stage conduction* through the copper, not simply additional resistive power dissipated in the tabs.

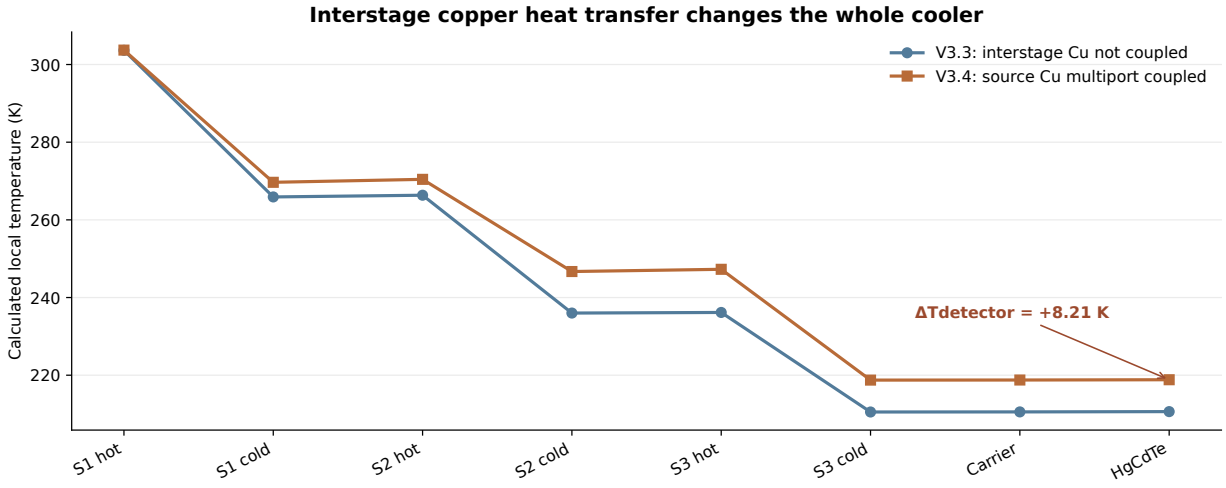


Figure 1: Solved source-ceramic stage and detector temperatures in V3.3 versus V3.4 at matched source assumptions. V3.4 adds a geometry-derived, finite-contact copper heat network; this is a conditional theoretical comparison, not measured or full 3D copper-metal fields.

1.1 Copper heat paths and their thermal consequences

The original CAD includes copper-foil tabs across electrical nets 58 and 80 and a stepped return at net 88. Their original contact sites on the three source ceramic planes are:

$$\text{net58}: \ 38 \to 75, \qquad \text{net80}: \ 79 \to 105, \qquad \text{net88}: \ 36 \to 74 \to 104.$$

The actual bridge foil width/thickness are 0.300/0.030 mm. Heat transfer uses the actual *center-to-center* solder-foot spans, not the full unsupported length of each tab. Net88 has an intermediate supported landing and therefore three separate thermal terminals. The model does not treat that landing as insulated.

For a foil segment i, j and finite contact c, i:

$$G_{ij} = k_{\mathrm{Cu}} A / L_{ij}, \qquad R_{ij} = \rho_{\mathrm{Cu}}(T_{ij}) L_{ij} / A, \qquad G_{c,i} = A_{c,i} / r''_{T,c},$$

where the provisional material laws use $k_{\mathrm{Cu}} = 385\ \mathrm{W/(m\,K)}$, $\rho_{\mathrm{Cu}}(T) = 1.724 \times 10^{-8}[1 + 0.0039(T - 293.15)]\ \Omega\,\mathrm{m}$, $r''_{T,c} = 10^{-6}\ \mathrm{m^2K/W}$, and $r''_{e,c} = 10^{-11}\ \Omega\,\mathrm{m^2}$. These are **selected engineering hypotheses**, not identified foil or solder-lot properties.

The node equation

$$\sum_j G_{ij}(T_i^{\mathrm{foil}} - T_j^{\mathrm{foil}}) + G_{c,i}(T_i^{\mathrm{foil}} - T_i^{\mathrm{ceramic}}) = P_{J,i}$$

sends local conduction and Joule heat to the original 3D ceramic-pad faces. The signed results below are reported as *heat into ceramic*; negative values indicate the foil removes heat:

Foil connection	Hot (mW)	Mid (mW)	Upper (mW)	Joule (mW)
Net 58	-54.594	+55.141	+0.000	0.547
Net 80	+0.000	-40.619	+41.115	0.496
Net 88	-53.881	+9.044	+45.875	1.038
ALL	-108.475	+23.566	+86.989	2.080

The copper network removes approximately 108.5 mW from the hot ceramic while depositing 87.0 mW in the upper ceramic. These are *redistributed* heat flows, not net external heat creation. Their stage total equals the copper circuit's 2.080 mW of electrical Joule dissipation.

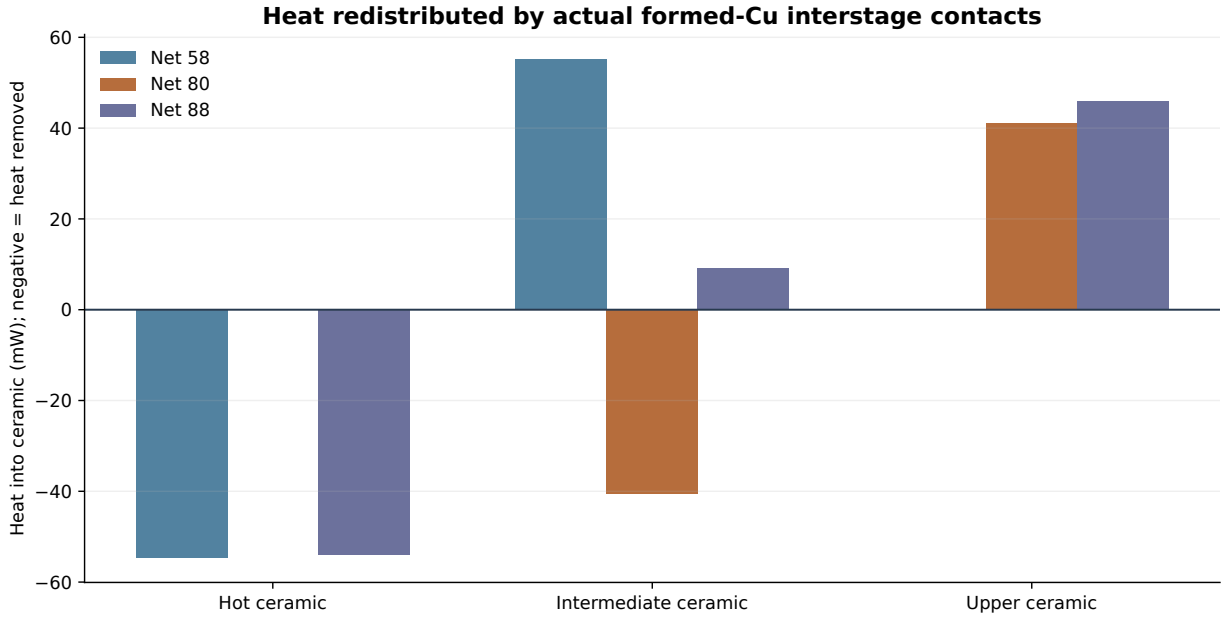


Figure 2: Original-CAD-face copper heat flows from accepted V3.4: individual networks at nets 58, 80 and 88. Positive denotes heat entering ceramic. Net88's middle contact has its own nonzero heat flux.

1.2 Resulting three-stage heat pumping and numerical checks

TEC stage	Number of legs	Q_c (mW)	Q_h (mW)	Voltage (V)
1	58	287.265	709.342	1.055194
2	22	136.178	260.411	0.310583
3	8	4.910	46.991	0.105201

The upper-stage hot boundary increases from 236.16 K (V3.3) to 247.27 K (V3.4). The stage therefore operates across a materially altered thermal environment while still providing approximately 4.910 mW of cold-side heat pumping.

Two earlier fixed-point attempts remain archived as **STOP**; the second passed a coarse pad-temperature threshold but failed the final energy test. The accepted third attempt meets the tighter 3×10^{-7} K temperature-update criterion after **19** outer steps, without changing the source physics:

Independent engineering check	Absolute residual
Final pad update (K)	1.399e-07
Independent stage heat residual (W)	7.959e-10
Global energy residual (W)	6.775e-10
Copper Joule/heat mismatch (W)	8.405e-16

A separately implemented read-only audit reconstructs copper Joule, all three stage heat balances and the hot-reservoir balance from the saved original-face and node data, without rerunning the coupled PDE. The 3D FEM still solves *only the ceramics*; the copper model remains a source-connected 1D heat/electric network with assumed contacts. These residuals establish internal conservation, not an experimental uncertainty interval. The principal next uncertainty is whether actual copper conductivity, foil/joint resistance, plated sheet spreading and geometry tolerances preserve the calculated 8.2 K shift.

2 Device construction and thermal architecture

Four nominal ceramic plates have square footprints 6.4, 5.2, 3.8 and 2.4 mm, each 0.5 mm thick. Eighty-eight P/N legs are 0.4 by 0.4 by 0.5 mm each, grouped into 58, 22 and 8 legs from hot to cold. The TEC supply is connected to header pins 1 and 9; the HgCdTe photoconductor has separate bias contacts on pins 7 and 8. The gold-colored contact patterns on the ceramic are internal series metallization and solder lands, *not* the external positive and negative power terminals. The exported reference contains 824 CAD solids and 176 solder joints, 89 insulated TEC electrical nets, and formed internal metal links. The original long, unsupported copper routing was rejected. The replacement has formed Cu-family foil tabs across nets 58 and 80, plus an extended three-foot return net 88 with a real middle thermal landing. Its design uses patterned ceramic metallization and finite solder footprints. The selected foil section is 0.300 by 0.030 mm; its bend radius is 0.075 mm; finished foot joints are specified at 20 plus/minus 5 micrometres. No measured current rating or fatigue qualification follows.

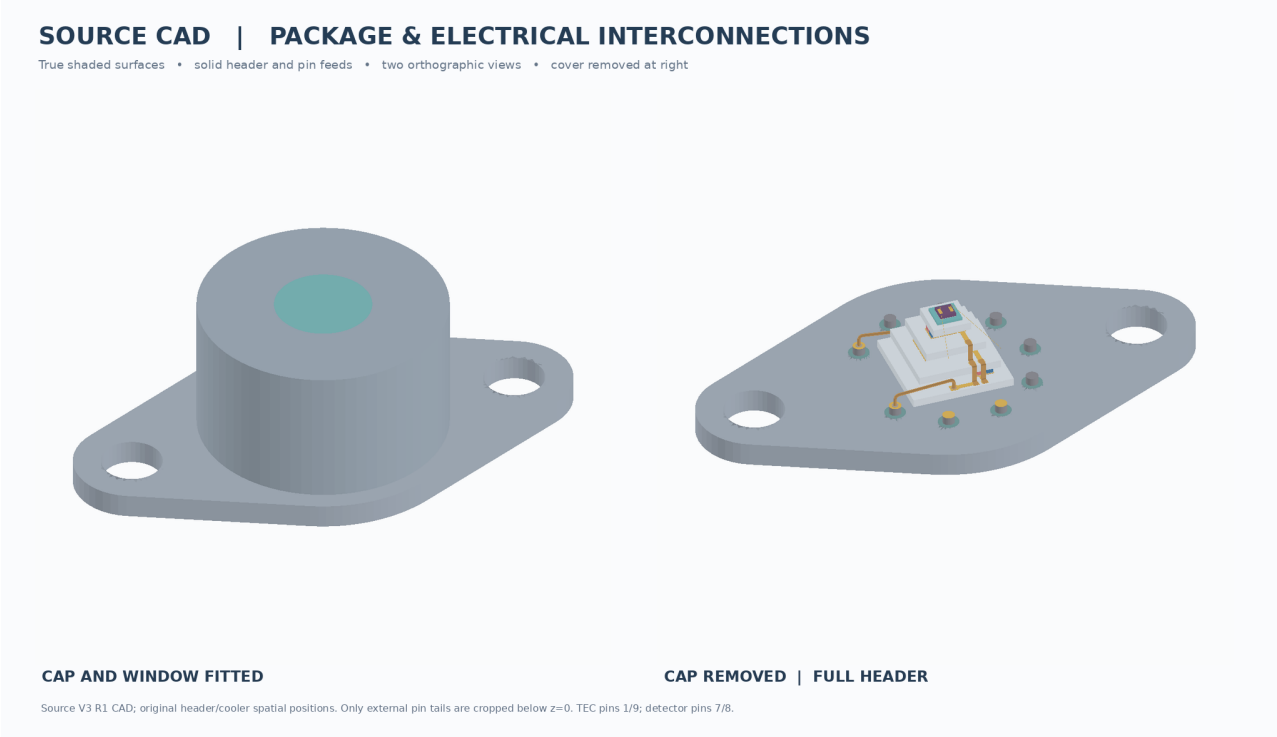


Figure 3: Physical arrangement of the proposed packaged HgCdTe device. Left: closed metal cap and optical window above the nine-pin header. Right: identical source-derived assembly with cover removed, exposing the thermoelectric stack, foil connections, TEC supply leads, and separate detector wiring. The external lead tails below the header are cropped for clarity; the mechanical assembly is synthetic, not a manufactured module.

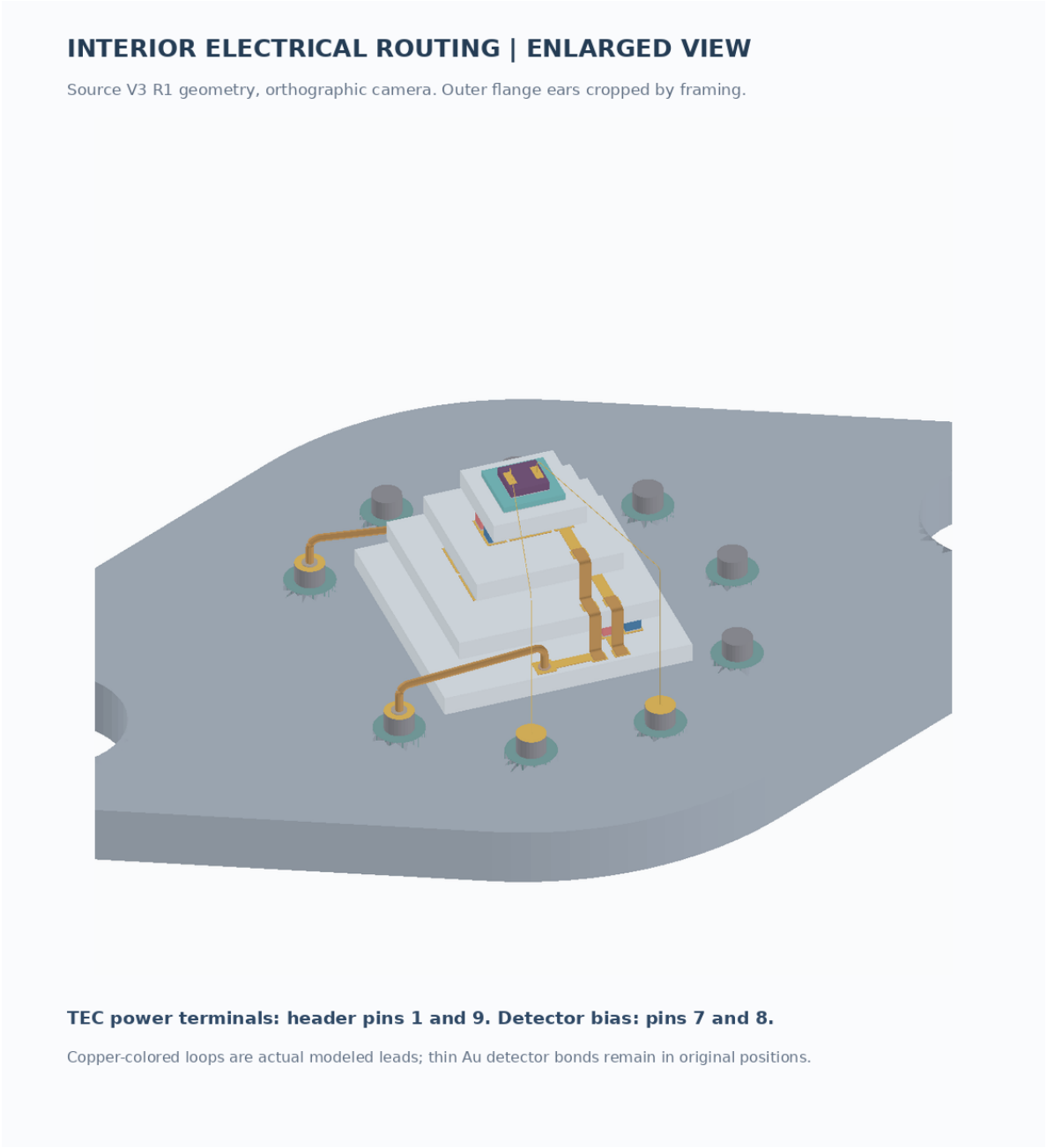


Figure 4: Enlarged source-based package interior: thermoelectric stack, copper supply leads connected to header pins 1 and 9, and separate gold detector bonds connected to pins 7 and 8. Visible patterned conductors on the ceramic are internal series electrodes and solder/contact lands. The outer flange ears are cropped *only in this inspection view.*

3 Cooler layout and supported interstage connections

The shaded views below tessellate the original 824-body source geometry using a true per-pixel depth buffer. The previous mean-face-depth renderer wrongly displayed hidden metallization through ceramic solids. This rendering is neither a reconstructed cooler nor an exploded assembly. All plates, P/N legs and analytically bent copper foil links retain the source positions, widths, heights, polarities and original path radii. Compared with the wireframe, the opaque solid bodies make the differences from the rejected long copper-post construction readily visible.

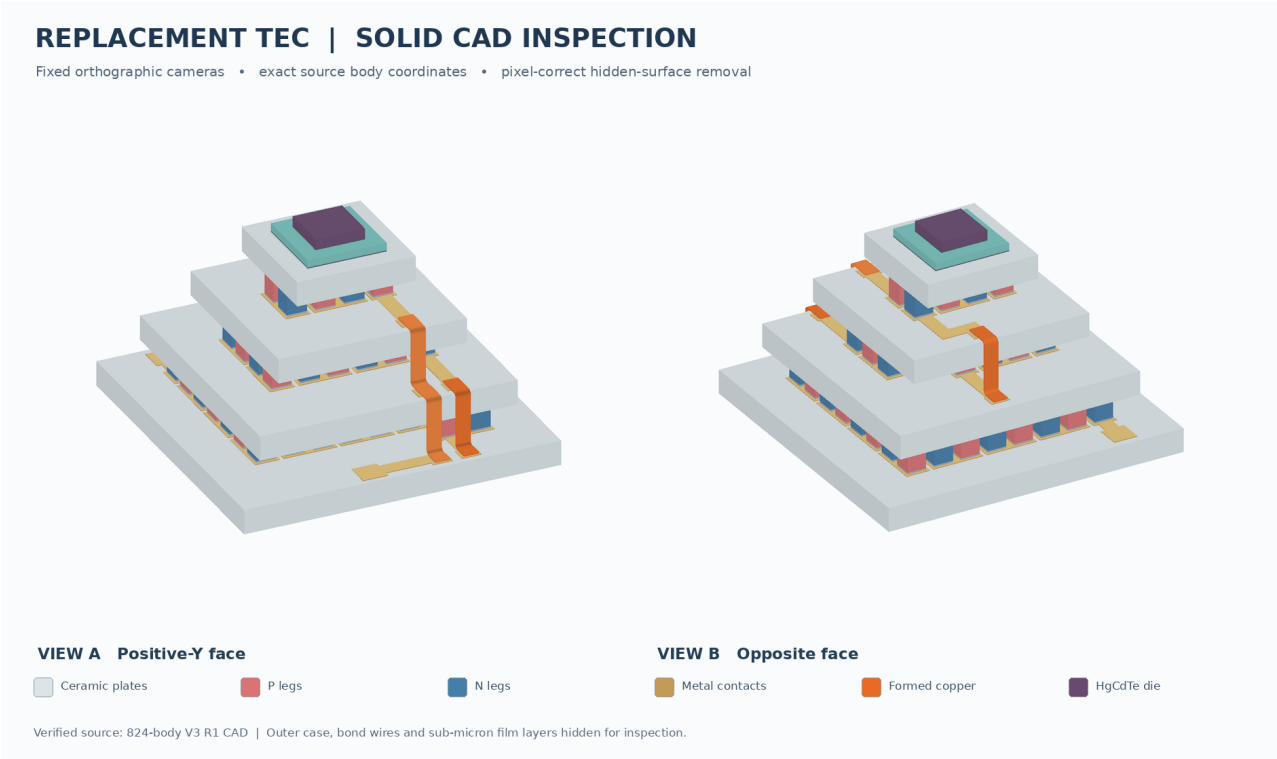


Figure 5: Correctly depth-occluded replacement geometry from two orthographic views: four ceramic stages, 58/22/8 P/N legs, visible exposed metallization only, and actual formed copper-foil connections. The external case, leads and very thin buried film layers are intentionally omitted. Original analytic foil-bend paths are sampled for shaded display, without relocating any CAD features.

3.1 Actual interstage foil cross-sections

The design uses two short supported formed copper-foil stage links (nets 58 and 80), and a longer stepped return (net 88). The last has two vertical steps and three physical feet, including an intermediate thermal landing. Unlike a generic connecting rod, its cross-section has nominal width 0.300 mm, thickness 0.030 mm and bend radius 0.075 mm. The foil is represented at actual thickness, not enlarged for visibility.

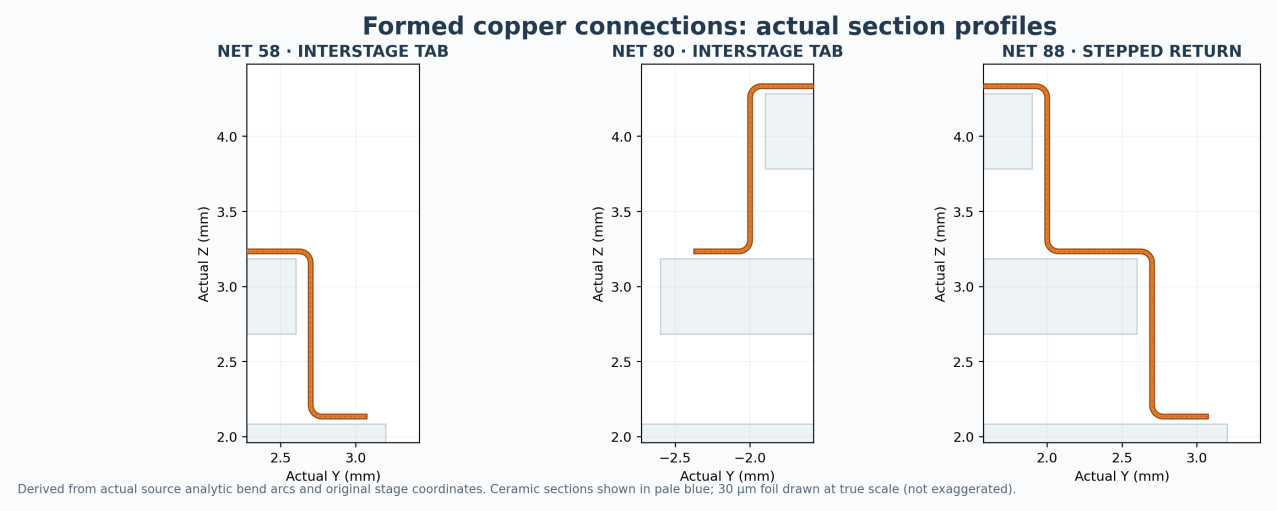


Figure 6: Y–Z sections reconstructed from the source analytic straight and circular-arc geometry at original coordinates, without exploded offsets. Pale blue regions indicate where ceramic cross-sections intersect each foil plane. The 30-micrometre copper thickness is drawn at true scale.

This inspection demonstrates the geometry change, but does not resolve missing formed-tab Joule heating, interface thermal resistance, electrothermal current crowding or interstage thermal bypass. Those remain necessary before the V3.3 detector-temperature estimate can qualify as a prediction for the assembled design.

4 Electrical routing and interstage heat leakage

The CAD conductor graph verifies all 88 legs as articulation vertices in the pin-1-to-pin-9 series circuit: removing any one leg disconnects the two terminals. Native transitions pass from P leg 57 to N leg 58 through tab 58, and from P79 to N80 through tab 80. The final leg P87 is connected to pin 9 through stepped return 88. These checks prove intended topology but not the temperature-dependent electrical or thermal properties of the connections. For a uniform slender conductor with temperature-dependent material properties,

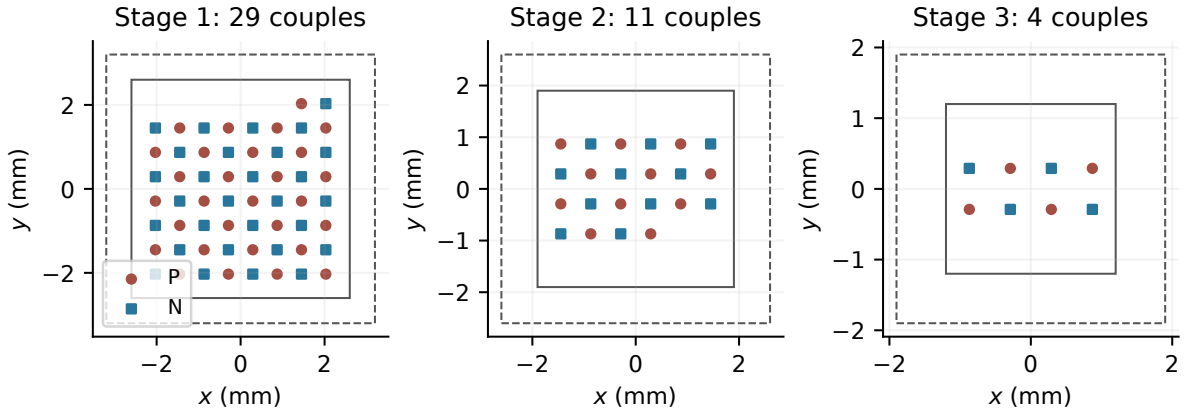


Figure 7: Actual saved stage-specific P/N leg centroids and plate footprints. This verifies the source positions used in the working hybrid; it does not depict a solved copper-tab thermal field.

$$R_e = \int_0^L \frac{\rho(T,s)}{A(s)}\,ds, \qquad G_{\text{th}} = \left[\int_0^L \frac{ds}{k(T,s)A(s)}\right]^{-1}, \quad P_J = I^2 R_e.$$

The registered neutral path lengths are 1.96062 mm (net58), 1.97062 mm (net80), and 3.70624 mm (return88). Using a uniform 0.300 by 0.030 mm section, illustrative $\rho_{\text{Cu}} = 1/(58\times10^6)\,\Omega\text{m}$ and $k_{\text{Cu}} = 385\,\text{W/(m K)}$ yields the approximate geometry-only screen below. The competing design goals are explicit: reducing electrical resistance

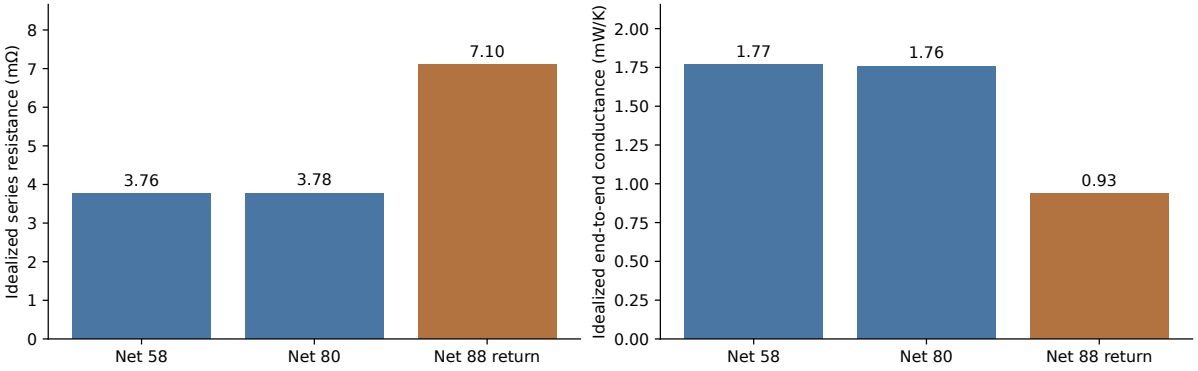


Figure 8: Derived idealized uniform-bar electrical and thermal copper coefficients. Actual formed-tab bends, patterned multilayer film, contact drops, variable temperatures, Joule partition and intermediate return landing are not resolved. The return is not correctly represented by a single two-terminal thermal conductance.

through a larger cross-section raises passive thermal coupling. A physical successor must solve or independently validate the full nonlocal stage-to-stage multiport heat and current network, including the return's intermediate land.

5 Geometric fidelity and remaining discretization risk

The first full R3.8 TetGen volume candidate contained 1,043,314 tetrahedra over 430 material regions, with all 320,522 registered facets and 378 finite-contact witnesses retained. R3.9 applied 33,195 conservative 3-to-2 internal connectivity operations and reduced the mesh to 1,010,119 tetrahedra with unchanged original surface

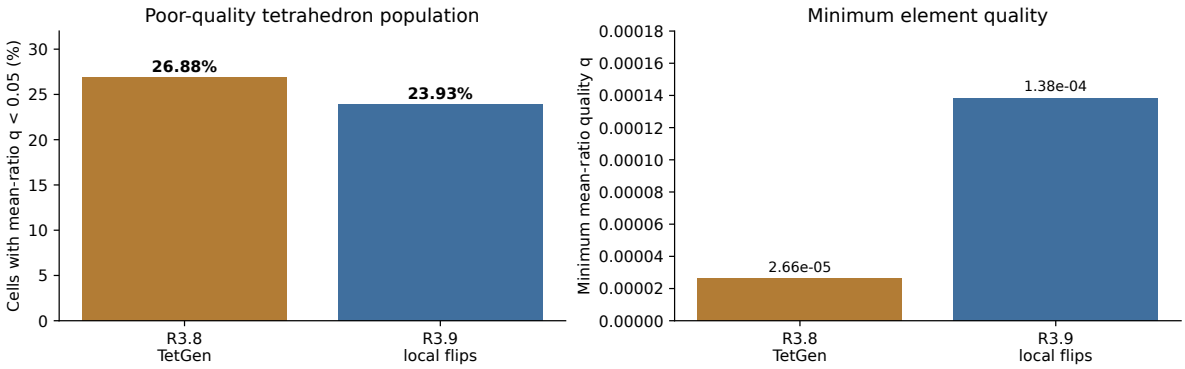


Figure 9: Independent archived global mesh-quality measurements. The 3-to-2 flips improved the poor-cell population, but the minimum quality and remaining weak cells prevent assuming mesh-converged coupled FEM.

facets and regional volumes. R3.9 retains 241,754 tetrahedra with mean-ratio quality below 0.05 (23.93 percent) and a minimum of 1.38495×10^{-4}. Thus no global coupled 430-domain electrical and thermal PDE is released. Historical full-solid electrical Gate B is FAIL/STOP; the first source N/P pair had eight raw gradient-current failures; the prospective P2 and global C/D computations are not qualified. Good energy closure in a separate hybrid model does not erase these original raw gates.

6 Thermoelectric material behavior and heat pumping

Each of the 88 axial P/N legs uses a signed 0.4 A series current and 32 longitudinal intervals. With z directed hot-to-cold, $S(T)$ Seebeck coefficient, $\rho(T)$ resistivity, $k(T)$ conductivity, and signed current density J, the constitutive conservation model is

$$q = S(T)TJ - k(T)\frac{dT}{dz}, \qquad \frac{dq}{dz} = J\left[\rho(T)J + S(T)\frac{dT}{dz}\right], \qquad \frac{dV}{dz} = \rho(T)J + S(T)\frac{dT}{dz}.$$

Midpoint constitutive evaluation and damped Newton iterations enforce thermal and electrical conservation at local temperatures, including the boundary identity $Q_h - Q_c = IV_{\text{leg}}$. The material source is a 14-knot figure digitization of research P-HE and N-HE-1 specimens over 40–300 K. Above 300 K the code uses an *unmeasured* one-quarter final-segment slope continuation; this is not a production material characterization.

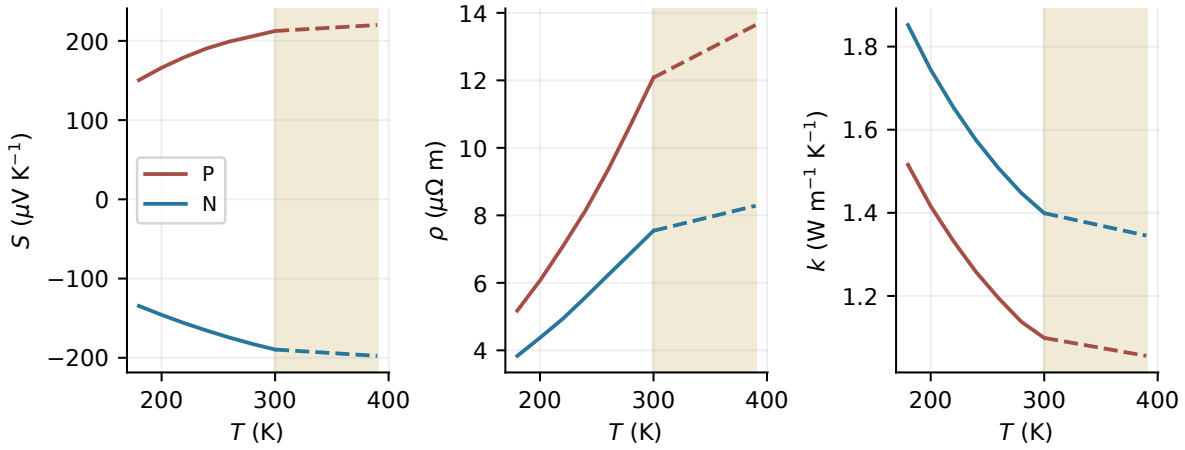


Figure 10: Source-derived P/N research analog properties. The shaded/dashed extension beyond 300 K is explicitly assumed; 58 hot-stage leg contacts exceed the research marker range.

7 Heat spreading through ceramic substrates

Each ceramic independently solves

$$-\nabla \cdot (k_c \nabla T) = 0, \qquad K_{ab} = \sum_e \int_{\Omega_e} k_c \nabla N_a \cdot \nabla N_b \, d\Omega, \quad b_a = \sum_f \int_{A_f} N_a q_f \, dA,$$

with $k_c = 30\,\mathrm{W/(m\,K)}$ a selected constant-property alumina analogue. Original face-specific P/N leg heat and assumed joint/cavity contributions are distributed over native TRI3 pad faces; source-local pad means feed back into leg end temperatures. Four archived source meshes contain 4,936 / 4,085 / 1,986 / 784 TET4 elements, totaling 11,791 tetrahedra. The four fields share their temperature gauge and stage heat exchange with the

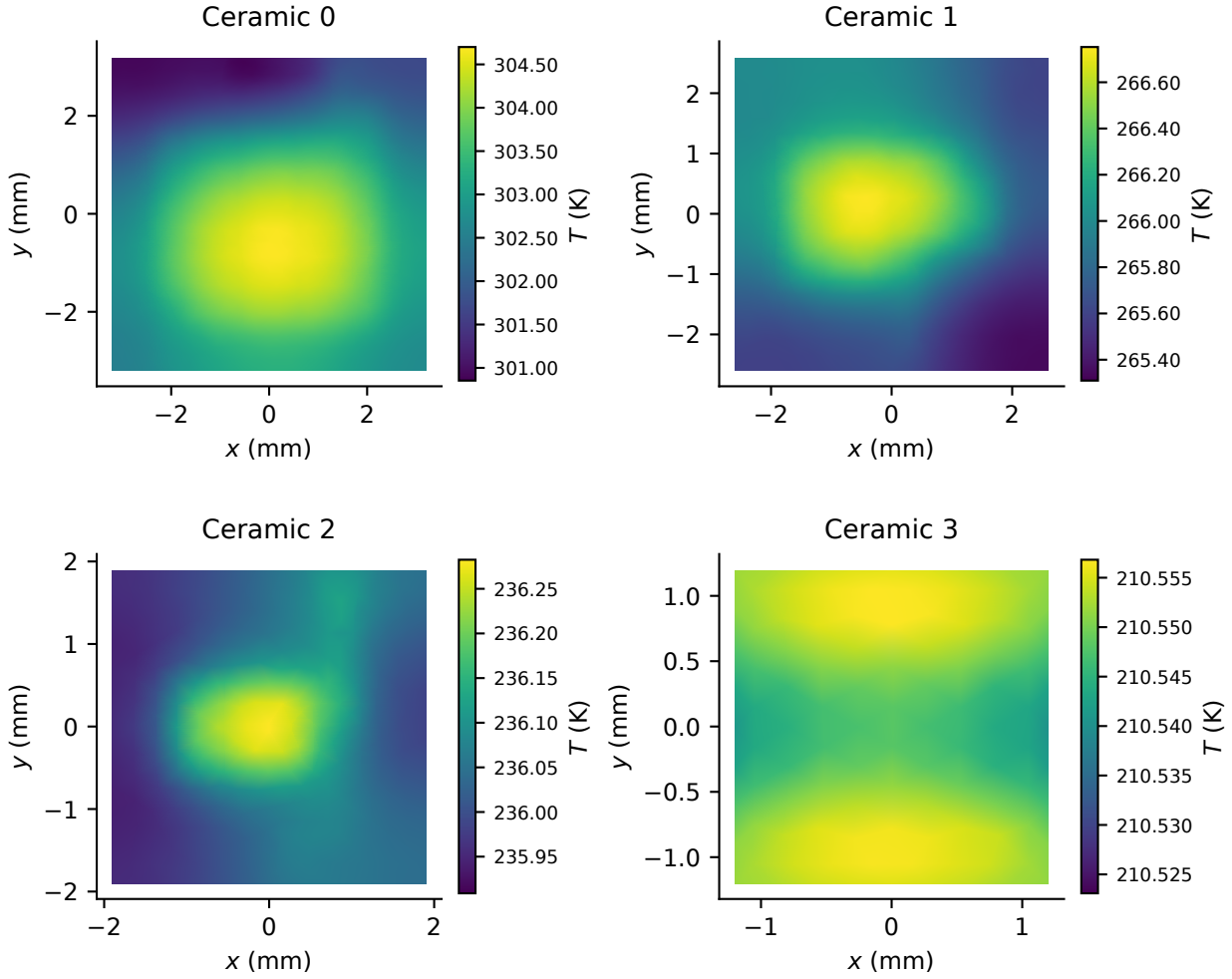


Figure 11: Real source-tetrahedron top-surface triangulations and solved nodal temperature fields from the original saved four ceramic archives. This is authentic spatial FEM evidence rather than a rendered approximation of the uncomputed full global package.

nonlocal reduced network. This is a single-mesh spatial result; no independent mesh-converged thermal accuracy bound is available.

8 Electrical contacts, solder joints and dissipative heating

The current synthetic electrical model assumes 176 solder bodies of 35 micrometres thickness and 0.16 mm^2 area, with assumed SnSb resistivity $1.7 \times 10^{-7}\ \Omega\,\mathrm{m}$ and 352 assumed metal-interface specific resistances $r_c'' = 10^{-11}\ \Omega\,\mathrm{m}^2$. For each joint the resistance is

$$R_j = \rho_{\mathrm{SnSb}} \frac{L_j}{A_j} + 2\frac{r_c''}{A_j}, \quad R_{\mathrm{all}} = 0.028545\ \Omega, \quad I^2 R_{\mathrm{all}} = 4.56720\ \mathrm{mW}.$$

The implementation places 25.950 micowatts per source joint into its neighboring ceramic pad. This assumes the entire joint Joule heat reaches a ceramic and omits finite thermal-contact drops, interfacial thermopower and nonlocal copper-network heating. The four ceramic applied-joint terms, in mW, are [1.50510, 2.07600, 0.77850, 0.20760]. External supply Cu wires and detector Au bonds are separate effective electrical/thermal links. *They must not be mistaken for a resolved calculation of the newly formed tabs or return conductor.*

9 Package parasitic heat loads and thermal rejection

The reference forward case imposes a 300 K cap/window/ambient thermal reservoir, a 100 Pa krypton environment, an effective 3.0 mm gas gap, accommodation 0.8, and nominal gas conductivity knots $k_{\mathrm{Kr}}(200, 300, 400\,\mathrm{K}) = (0.0064, 0.0095, 0.0122)\ \mathrm{W/(m\,K)}$. Gray radiation uses assumed emissivities $\epsilon_{\mathrm{cap}} = 0.20$, $\epsilon_{\mathrm{ceramic}} = 0.80$, and $\epsilon_{\mathrm{window}} = 0.10$, with selected effective view factors. The source effective exposed surface areas are [12.60, 8.68, 4.76, 1.00] mm^2. The generic exchange terms are

$$Q_{\mathrm{gas},i} = G_{\mathrm{eff,i}}(p,T)(T_\infty - T_i), \qquad Q_{\mathrm{rad},i} = \epsilon_{\mathrm{eff},i}\sigma A_i F_i (T_\infty^4 - T_i^4).$$

These are *effective engineering closures*, not cavity fluid dynamics, true free-molecular transport, exact exposed-face loads, spectral window modeling or full radiosity.

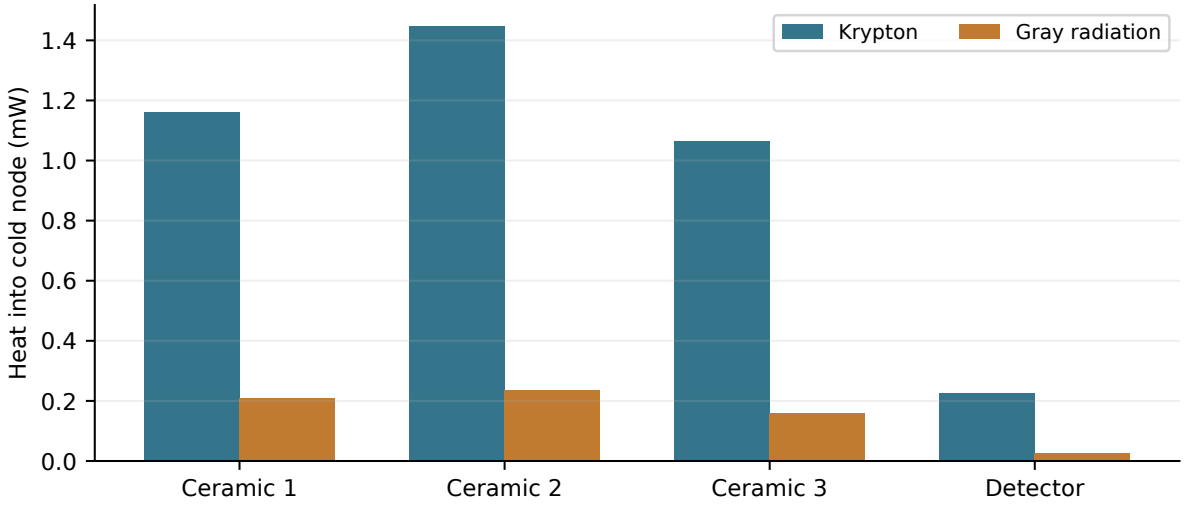


Figure 12: Archived V3.3 cavity contributions: total gas 3.892553 mW, gray radiation 0.624076 mW, combined heat into cold components 4.516629 mW. The reservoir carries an equal and opposite signed load.

Cold destination	Kr gas (mW)	Gray rad. (mW)	Total (mW)
Ceramic C1	1.15901	0.20874	1.36775
Ceramic C2	1.44533	0.23389	1.67923
Ceramic C3	1.06469	0.15770	1.22239
Detector	0.22352	0.02375	0.24726
TOTAL	3.89255	0.62408	4.51663

An effective header/TIM/radiator resistance of approximately 4.8334 K/W carries heat to the warm sink. Neither mount pressure nor material lot is an experimental datum. The cavity load is redistributed to native *upper metal-pad* sources in the current model; thermal gradients on exposed bare ceramic faces are not thereby demonstrated.

10 Predicted detector temperature under enclosure loading

The following **historical baseline** V3.2 versus V3.3 comparison (prior to adding V3.4 interstage copper) uses the same 88-leg source geometry, 11,791 ceramic tetrahedra, 0.4 A input, 300 K reservoir, 2 mW detector load and assumed material/contact parameters. The cavity raises die temperature 0.637233 K, while the nonlinear terminal voltage changes slightly due to changed operating gradients.

Metric	V3.2 (cavity off)	V3.3 (cavity on)
Detector temperature (K)	209.987981	210.625214
Terminal voltage (V)	1.524071	1.522462
Electrical input (W)	0.609628	0.608985
Bath rejection (W)	0.590323	0.594044
Pressure (Pa)	0 (cavity off)	100 (Kr)

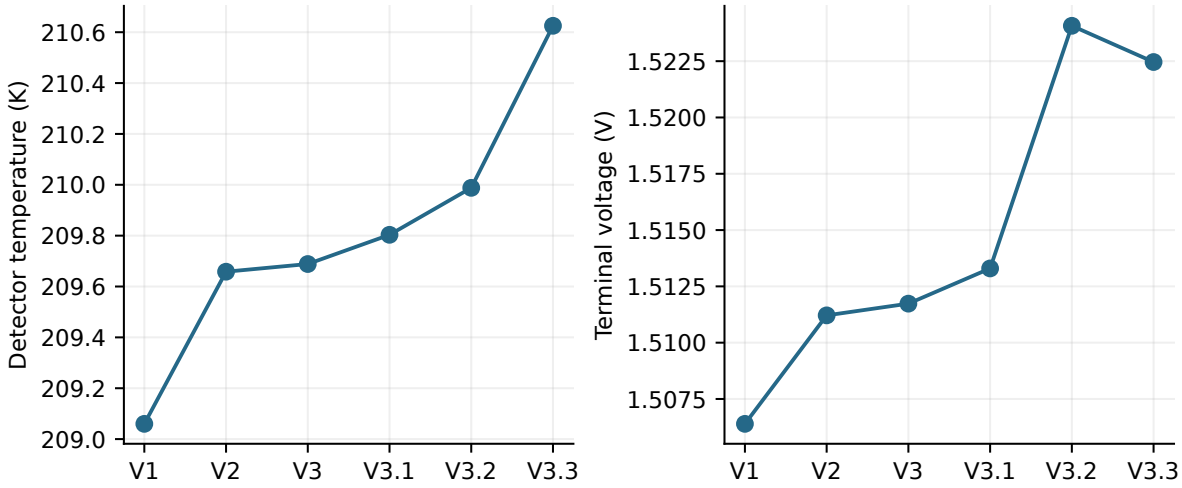


Figure 13: Recorded V1 to V3.3 complete reference-model milestones, all evaluated at the nominal current. These points describe changing model fidelity, not successive mesh convergence or independent measurements.

11 Predicted stage temperatures, heat flow, and electrical input

Stage	Legs	Cold flux (mW)	Hot flux (mW)	V	Joule (mW)
1	58	179.9602	614.7396	1.086948	262.732
2	22	46.1636	176.5165	0.325882	81.390
3	8	5.1576	43.7058	0.096371	24.642

Coupled node	Temperature (K)
Stage0 Hot	303.65655
Stage0 Cold	265.90329
Stage1 Hot	266.34287
Stage1 Cold	236.00259
Stage2 Hot	236.15716
Stage2 Cold	210.52765
Carrier	210.55066
Die	210.62521

The external terminal voltage includes additional lead and joint contributions beyond summed P/N voltage. Stage energies must account for signed electrical work, source losses, detector heat, cavity sources and external sink. Directly adding cavity milliwatts to input electrical power without solving the changed stage temperatures would be incorrect.

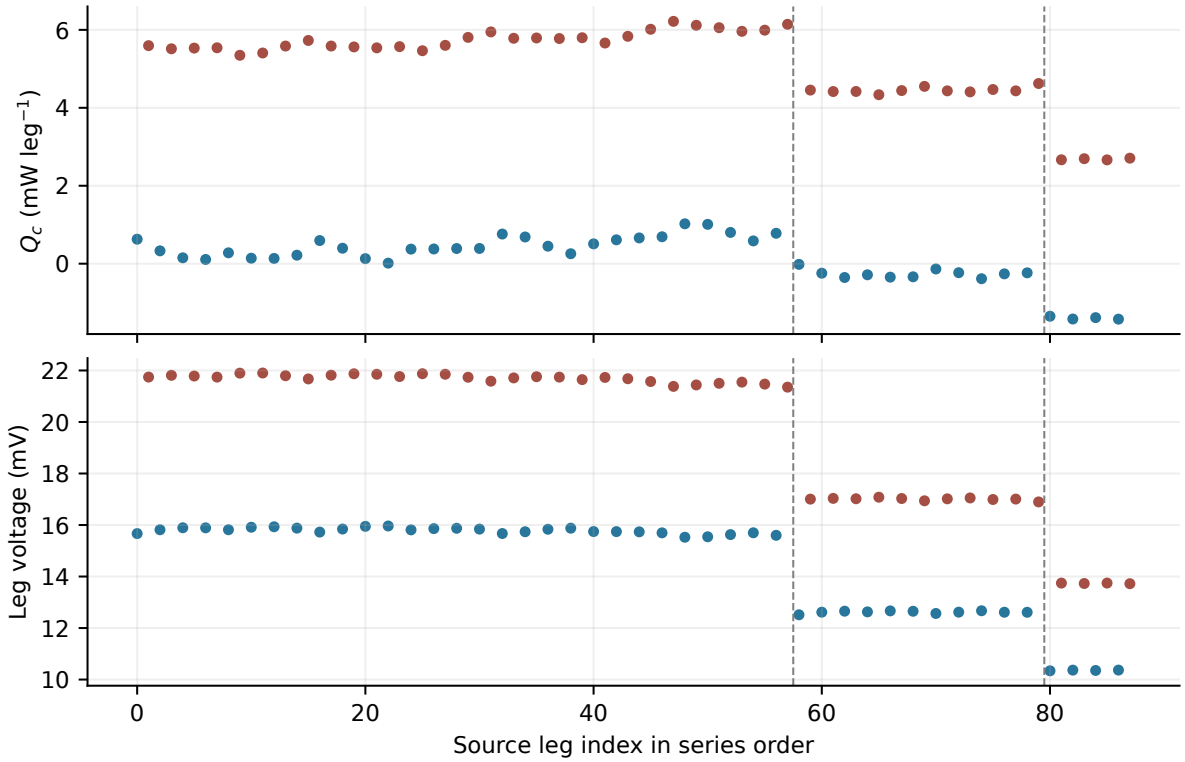


Figure 14: Real saved individual-leg cold-side heat and terminal-voltage contributions. Stage transitions occur after source leg indices 57 and 79. Source material laws and local ceramic pad temperatures determine leg-specific behavior.

12 Energy conservation and numerical credibility

For the **earlier V3.3 baseline only**, the solver used the previous V3.2 solution as a warm start and iterated five spatial leg/pad updates, stopping at a maximum pad temperature change 1.65345×10^{-5} K below its registered 2×10^{-5} K criterion. That number is a *numerical stopping threshold*, not a temperature accuracy bound.

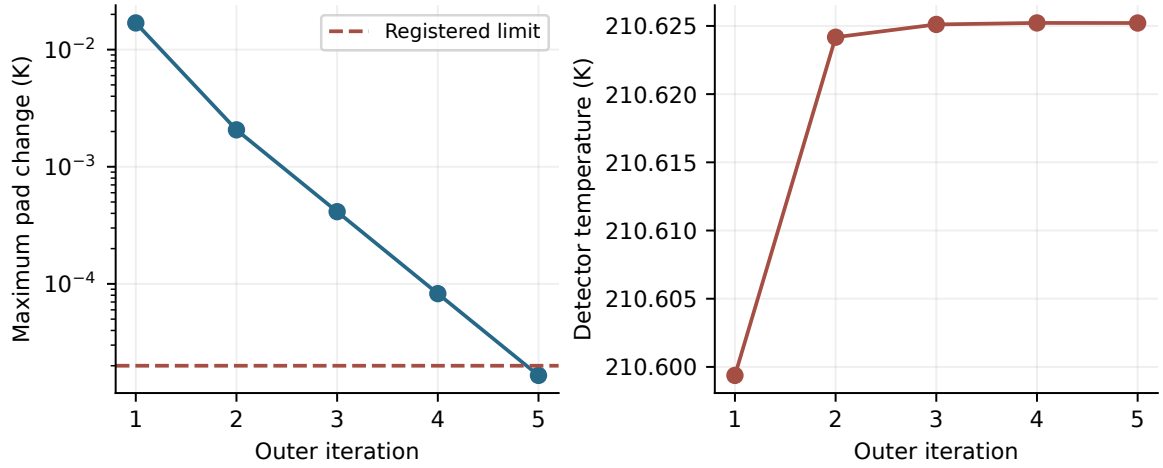


Figure 15: Actual saved spatial fixed-point iteration changes and detector temperature history. The last update meets the registered coarse convergence threshold; no mesh refinement or material uncertainty bound is inferred.

Iteration	Die T (K)	Max pad update (K)	Local flux mismatch (W)
1	210.599377	1.690e-02	3.914e-15
2	210.624173	2.066e-03	4.580e-16
3	210.625110	4.135e-04	1.478e-15
4	210.625220	8.254e-05	2.928e-15
5	210.625214	1.653e-05	5.503e-15

Metric	Absolute residual	Unit
Global energy	8.8674e-09	W
Worst stage balance	4.9061e-09	W
Ceramic pad heat	5.5025e-15	W
Worst individual leg	1.2364e-13	W
Final pad temperature change	1.6535e-05	K

An independent read-only field/load audit reconstructs the saved energy terms without rerunning the original PDE and finds a finite 2.535×10^{-8} W pad/leg update discrepancy. Numerical closure validates the *declared equations and discretization bookkeeping* at this operating point, not prediction fidelity of the as-built cooler.

13 Device engineering implications and next experiment

The physical cooler is now represented by a functioning complete thermal/electrical model including a first conservative three-branch copper network tied to original ceramic pads. The corrected provisional detector operating point is 218.833 K rather than 210.625 K. It is *not* necessary to restart cooler CAD development. However, full three-dimensional copper/metal fields and actual thermal-contact characterization are not yet available. The next practical milestone is to quantify uncertainty in the *already integrated* multiport: copper conductivity and resistivity versus temperature, contact thermal/electrical impedances, foil thickness and metallization-sheet heat spreading. Use a small, bounded source-informed screening study to select the few conditions that justify complete nonlinear recomputations. Full 3D copper FEM may be justified if compact validated metal substructures cannot bound current crowding or nonlocal heat flow. The older 38-page Revision 3 is preserved because of its unusually useful FEM methodology and publication-level plots, *not* because it describes the current metal layout. The 43-page interim extension is historical, not the latest design reference.

14 Model provenance and numerical evidence hierarchy

Artifact	Subject	Verified status
Historical Rev3	Earlier 216-solid 28/12/4 cooler	Detailed field method; raw-gradient release failed
Replacement V3 R1	824 physical STEP bodies, formed tabs and joints	Conditional CAD design acceptance
R3.9 geometry	430 material regions, 2,440 CAD faces	Geometric mesh audit; conditioning unqualified
V3.3 thermal baseline	88 nonlinear legs and four source 3D ceramic fields	Accepted reference without formed Cu network
V3.4 coupled source Cu	Same 88 legs / source ceramics with formed-foil thermal multiport	One accepted conditional result; 3D metal and empirical contact laws unqualified

The cooler envelope is informed by TEC Microsystems MC04 family reference geometry; the package by a separate VIGO nine-pin TO-66 drawing. They are not identified as a commercially fabricated composite. The 29/11/4 stage split, material grades, solder properties and fabricated metal details are source-inspired or chosen synthetic hypotheses, *not* proprietary OEM internals.

15 Physical assumptions and reproducibility specification

An independent implementation should use the four 0.5 mm-thick 6.4/5.2/3.8/2.4 mm ceramics, the Appendix B 88 leg centers and polarities, the 29/11/4-couple electrical series order and the source-selected 0.4 by 0.4 by 0.5 mm P/N geometry. Apply all six 14-knot P/N research-material curves in Appendix A; interpolate within the experimental range and explicitly impose the unmeasured quarter-final-segment-slope continuation above 300 K. Solve 88 individually signed 32-interval nonlinear axial thermoelectric legs and four source-geometry P1 ceramic steady conduction systems using local native TRI3 heat-face integrals. Set the declared assumed ceramic conductivity to 30 W/(m K), add the nominal 176 SnSb joints and 352 assumed electrical contacts, Cu and Au external leads, 2 mW detector heat, the stated finite mounting resistance and the effective 100 Pa Kr/gray radiation closure. For V3.4, include the three finite-contact Cu-foil connections from native faces 38/75, 79/105 and 36/74/104 and iterate to a maximum local pad change below 0.0000003 K and independently check all signed stage and total heat balances. The archived Appendix C face centroids, areas and flux totals permit a comparable heat-integral replication, not byte-exact nodal temperatures. Bitwise reproduction requires original tetrahedral connectivity, exact face polygon vertices, quadrature and solver conventions. No unpublished real-world detector measurements are claimed.

16 Scientific and engineering references

1. TEC Microsystems, three-stage 3MC04-044-05 cooler family datasheet (2019), external envelope and thermoelectric reference. Its hidden internal routing is unknown.

2. VIGO Photonics, 3TE TO-66 nine-pin housing reference, Rev. 9 (2024). The cooler and housing are independent reference products.

3. J. Zhang et al., low-temperature bismuth telluride thermoelectric measurement and error-analysis study, *Chinese Physics Letters* 42, 056304 (2025). DOI 10.1088/0256-307X/42/5/056304. Source P-HE and N-HE-1 specimens are not the actual installed module.

4. M. L. Huber, gas thermal conductivity tabulations (NIST/CRC, 2019): krypton material reference, not experimentally measured gap conduction.

5. TEC Microsystems and Kryotherm multistage construction drawings and photographs; Coherent mounting guidelines. These establish process families, not qualification of the selected synthetic formed-tab design.

A Complete 14-knot P and N transport source tables

The following 84 values comprise six figure-digitized research-specimen curves. They do not establish actual commercial material properties. The assumed extension beyond 300 K is

$$f(T) = f(300) + 0.25\,(T - 300)\frac{f(300) - f(280)}{20}, \qquad f \in \{S, \rho, k\}.$$

T K	S_P μV/K	ρ_P $\mu\Omega$ m	k_P W/(m K)	S_N μV/K	ρ_N $\mu\Omega$ m	k_N W/(m K)
40	31.477086	1.226120	2.776424	-39.762194	0.837035	3.928920
60	44.991486	1.550968	2.488935	-53.616397	1.156199	3.246955
80	60.679339	1.955036	2.260290	-67.469437	1.522950	2.816874
100	77.996604	2.446265	2.072406	-81.936140	1.920207	2.536709
120	96.944423	3.004830	1.909416	-95.992742	2.354830	2.310364
140	115.755946	3.567364	1.764537	-109.847998	2.820018	2.129254
160	133.684757	4.343818	1.630978	-122.547039	3.319204	1.973635
180	150.458980	5.175731	1.515526	-134.499967	3.835302	1.851971
200	166.078757	6.067058	1.415926	-145.840430	4.371811	1.744414
220	179.321421	7.073284	1.332165	-156.231255	4.932113	1.653884
240	190.594863	8.138932	1.257469	-165.737958	5.570441	1.574639
260	199.219885	9.351143	1.194083	-174.568277	6.232621	1.506729
280	205.739245	10.694094	1.137484	-182.445326	6.884545	1.447332
300	212.462550	12.080611	1.099006	-189.575244	7.543277	1.399227

B All original 88 P/N leg centers

Each leg has 0.1600 mm^2 area and 0.5000 mm axial length; stage index 0 is the hottest. All coordinates refer to the original selected synthetic CAD x/y frame in mm.

Series index	Stage	P/N	x (mm)	y (mm)
0	0	N	-2.030000	-2.030000
1	0	P	-1.450000	-2.030000
2	0	N	-0.870000	-2.030000
3	0	P	-0.290000	-2.030000
4	0	N	0.290000	-2.030000
5	0	P	0.870000	-2.030000
6	0	N	1.450000	-2.030000
7	0	P	2.030000	-2.030000
8	0	N	2.030000	-1.450000
9	0	P	1.450000	-1.450000
10	0	N	0.870000	-1.450000
11	0	P	0.290000	-1.450000
12	0	N	-0.290000	-1.450000
13	0	P	-0.870000	-1.450000
14	0	N	-1.450000	-1.450000
15	0	P	-2.030000	-1.450000
16	0	N	-2.030000	-0.870000
17	0	P	-1.450000	-0.870000
18	0	N	-0.870000	-0.870000
19	0	P	-0.290000	-0.870000
20	0	N	0.290000	-0.870000
21	0	P	0.870000	-0.870000
22	0	N	1.450000	-0.870000
23	0	P	2.030000	-0.870000
24	0	N	2.030000	-0.290000
25	0	P	1.450000	-0.290000
26	0	N	0.870000	-0.290000
27	0	P	0.290000	-0.290000
28	0	N	-0.290000	-0.290000
29	0	P	-0.870000	-0.290000
30	0	N	-1.450000	-0.290000
31	0	P	-2.030000	-0.290000
32	0	N	-2.030000	0.290000
33	0	P	-1.450000	0.290000
34	0	N	-0.870000	0.290000
35	0	P	-0.290000	0.290000
36	0	N	0.290000	0.290000
37	0	P	0.870000	0.290000
38	0	N	1.450000	0.290000
39	0	P	2.030000	0.290000
40	0	N	2.030000	0.870000
41	0	P	1.450000	0.870000
42	0	N	0.870000	0.870000
43	0	P	0.290000	0.870000
44	0	N	-0.290000	0.870000
45	0	P	-0.870000	0.870000
46	0	N	-1.450000	0.870000
47	0	P	-2.030000	0.870000
48	0	N	-2.030000	1.450000
49	0	P	-1.450000	1.450000
50	0	N	-0.870000	1.450000
51	0	P	-0.290000	1.450000
52	0	N	0.290000	1.450000
53	0	P	0.870000	1.450000
54	0	N	1.450000	1.450000
55	0	P	2.030000	1.450000
56	0	N	2.030000	2.030000
57	0	P	1.450000	2.030000
58	1	N	1.450000	0.870000
59	1	P	0.870000	0.870000
60	1	N	0.290000	0.870000
61	1	P	-0.290000	0.870000
62	1	N	-0.870000	0.870000
63	1	P	-1.450000	0.870000
64	1	N	-1.450000	0.290000
65	1	P	-0.870000	0.290000
66	1	N	-0.290000	0.290000
67	1	P	0.290000	0.290000
68	1	N	0.870000	0.290000
69	1	P	1.450000	0.290000
70	1	N	1.450000	-0.290000
71	1	P	0.870000	-0.290000
72	1	N	0.290000	-0.290000
73	1	P	-0.290000	-0.290000
74	1	N	-0.870000	-0.290000
75	1	P	-1.450000	-0.290000
76	1	N	-1.450000	-0.870000
77	1	P	-0.870000	-0.870000

Series index	Stage	P/N	x (mm)	y (mm)
78	1	N	-0.290000	-0.870000
79	1	P	0.290000	-0.870000
80	2	N	0.870000	-0.290000
81	2	P	0.290000	-0.290000
82	2	N	-0.290000	-0.290000
83	2	P	-0.870000	-0.290000
84	2	N	-0.870000	0.290000
85	2	P	-0.290000	0.290000
86	2	N	0.290000	0.290000
87	2	P	0.870000	0.290000

C Native ceramic-face geometric and solved heat ledger

For each original ceramic and native thermal face, the table lists source face ID, centroid (mm), original CAD area (mm^2) and the signed integrated solved heat *into* that ceramic (W). Contributions include local P/N transport, provisional joint heating, finite sink, detector and effective cavity loads when applicable. Exact face polygons are not fully specified by centroids and areas.

Cer.	Face ID	x	y	z	Area mm^2	Heat in (W)
0	6	0.000000	0.000000	1.584250	37.21000000	-5.9404375264e-01
0	8	-2.030000	-2.415000	2.084250	0.58100000	+6.9212449542e-03
0	9	-1.160000	-2.030000	2.084250	0.54000000	+2.1001118011e-02
0	10	-0.000000	-2.030000	2.084250	0.54000000	+2.0799385661e-02
0	11	-2.030000	-1.160000	2.084250	0.54000000	+2.1333430585e-02
0	12	-1.160000	-1.450000	2.084250	0.54000000	+2.0925106301e-02
0	13	-1.160000	-0.870000	2.084250	0.54000000	+2.1098181465e-02
0	14	-2.030000	-0.000000	2.084250	0.54000000	+2.1661444954e-02
0	15	-1.160000	-0.290000	2.084250	0.54000000	+2.1280527762e-02
0	16	-0.000000	-1.450000	2.084250	0.54000000	+2.0726535211e-02
0	17	0.000000	-0.870000	2.084250	0.54000000	+2.0872339885e-02
0	18	-0.000000	-0.290000	2.084250	0.54000000	+2.1135046258e-02
0	19	1.160000	-2.030000	2.084250	0.54000000	+2.0762274895e-02
0	20	2.030000	-1.740000	2.084250	0.54000000	+2.0894962316e-02
0	21	1.160000	-1.450000	2.084250	0.54000000	+2.0664089207e-02
0	22	1.160000	-0.870000	2.084250	0.54000000	+2.0730610371e-02
0	23	1.160000	-0.290000	2.084250	0.54000000	+2.0987932670e-02
0	24	2.030000	-0.580000	2.084250	0.54000000	+2.1025501033e-02
0	25	-1.160000	0.290000	2.084250	0.54000000	+2.1501510390e-02
0	26	-2.030000	1.160000	2.084250	0.54000000	+2.2056807823e-02
0	27	-1.160000	0.870000	2.084250	0.54000000	+2.1665224003e-02
0	28	-1.160000	1.450000	2.084250	0.54000000	+2.1972015528e-02
0	29	0.000000	0.290000	2.084250	0.54000000	+2.1329922044e-02
0	30	-0.000000	0.870000	2.084250	0.54000000	+2.1513476612e-02
0	31	0.000000	1.450000	2.084250	0.54000000	+2.1762448001e-02
0	32	1.160000	0.290000	2.084250	0.54000000	+2.1132384800e-02
0	33	2.030000	0.580000	2.084250	0.54000000	+2.1315494607e-02
0	34	1.160000	0.870000	2.084250	0.54000000	+2.1316483722e-02
0	35	1.160000	1.450000	2.084250	0.54000000	+2.1497446937e-02
0	36	0.088570	2.858995	2.084250	0.67225000	-2.2200926469e-02
0	37	2.030000	1.740000	2.084250	0.54000000	+2.1652376653e-02
0	38	1.450000	2.379622	2.084250	0.51850000	+1.4709356454e-02
1	44	-1.740000	-2.030000	2.684250	0.54000000	-6.1717204457e-03
1	45	-0.580000	-2.030000	2.684250	0.54000000	-5.7923394592e-03
1	46	0.580000	-2.030000	2.684250	0.54000000	-5.6313481447e-03
1	47	-1.740000	-1.450000	2.684250	0.54000000	-5.8945158835e-03
1	48	-1.740000	-0.870000	2.684250	0.54000000	-6.1272498337e-03
1	49	-0.580000	-1.450000	2.684250	0.54000000	-5.6677985879e-03
1	50	-0.580000	-0.870000	2.684250	0.54000000	-5.9073535857e-03
1	51	-1.740000	-0.290000	2.684250	0.54000000	-6.2861601984e-03
1	52	-1.740000	0.290000	2.684250	0.54000000	-6.4929448577e-03
1	53	-0.580000	-0.290000	2.684250	0.54000000	-6.1441766581e-03
1	54	0.580000	-1.450000	2.684250	0.54000000	-5.4949366062e-03
1	55	0.580000	-0.870000	2.684250	0.54000000	-5.6165637076e-03
1	56	0.580000	-0.290000	2.684250	0.54000000	-5.9284010470e-03
1	57	1.740000	-2.030000	2.684250	0.54000000	-5.5964450493e-03
1	58	1.740000	-1.450000	2.684250	0.54000000	-5.5750930796e-03
1	59	1.740000	-0.870000	2.684250	0.54000000	-5.5316729855e-03
1	60	1.740000	-0.290000	2.684250	0.54000000	-5.7862099653e-03
1	61	-0.580000	0.290000	2.684250	0.54000000	-6.4257386885e-03
1	62	-1.740000	0.870000	2.684250	0.54000000	-6.8576261766e-03
1	63	-1.740000	1.450000	2.684250	0.54000000	-7.0890067759e-03
1	64	-0.580000	0.870000	2.684250	0.54000000	-6.6249732312e-03
1	65	-0.580000	1.450000	2.684250	0.54000000	-7.0137372522e-03
1	66	0.580000	0.290000	2.684250	0.54000000	-6.1746390641e-03
1	67	0.580000	0.870000	2.684250	0.54000000	-6.3948651910e-03
1	68	0.580000	1.450000	2.684250	0.54000000	-6.7126784447e-03
1	69	1.740000	0.290000	2.684250	0.54000000	-6.0016216656e-03
1	70	1.740000	0.870000	2.684250	0.54000000	-6.1198269805e-03
1	71	1.740000	1.450000	2.684250	0.54000000	-6.5247521871e-03
1	72	1.740000	2.030000	2.684250	0.54000000	-6.8707555753e-03
1	74	0.870000	2.300000	3.184250	0.12000000	+2.3941388903e-05
1	75	1.450000	1.470862	3.184250	0.67900000	+5.1504788556e-03
1	76	-1.450000	-0.580000	3.184250	0.54000000	+1.6210670281e-02
1	77	-0.580000	-0.870000	3.184250	0.54000000	+1.6207954022e-02
1	78	-0.580000	-0.290000	3.184250	0.54000000	+1.6071664242e-02
1	79	0.290000	-1.435162	3.184250	0.65650000	+1.1539642240e-02
1	80	0.580000	-0.290000	3.184250	0.54000000	+1.6217492734e-02
1	81	1.450000	0.000000	3.184250	0.54000000	+1.6376311378e-02
1	82	-1.450000	0.580000	3.184250	0.54000000	+1.6153119097e-02
1	83	-0.580000	0.290000	3.184250	0.54000000	+1.6048683270e-02
1	84	-0.580000	0.870000	3.184250	0.54000000	+1.6098735692e-02
1	85	0.580000	0.290000	3.184250	0.54000000	+1.6138044094e-02
1	86	0.580000	0.870000	3.184250	0.54000000	+1.6218414034e-02

Cer.	Face ID	x	y	z	Area	Heat (W)
2	92	-1.160000	-0.870000	3.784250	0.54000000	-4.1229763184e-03
2	93	-1.160000	-0.290000	3.784250	0.54000000	-4.0325885849e-03
2	94	0.000000	-0.870000	3.784250	0.54000000	-4.3372223253e-03
2	95	-0.000000	-0.290000	3.784250	0.54000000	-4.1255544021e-03
2	96	1.160000	-0.290000	3.784250	0.54000000	-4.2477066490e-03
2	97	-1.160000	0.290000	3.784250	0.54000000	-3.9991911145e-03
2	98	-1.160000	0.870000	3.784250	0.54000000	-4.0132490723e-03
2	99	0.000000	0.290000	3.784250	0.54000000	-4.0452781812e-03
2	100	1.160000	0.290000	3.784250	0.54000000	-4.1617896577e-03
2	101	-0.000000	0.870000	3.784250	0.54000000	-4.1215621922e-03
2	102	1.160000	0.870000	3.784250	0.54000000	-4.3855401752e-03
2	104	0.870000	0.829062	4.284250	0.64000000	+8.5720985048e-03
2	105	0.692720	-0.929413	4.284250	0.81350000	+3.2580848072e-03
2	106	-0.000000	-0.290000	4.284250	0.54000000	+1.1238504436e-02
2	107	-0.870000	0.000000	4.284250	0.54000000	+1.1290111066e-02
2	108	0.000000	0.290000	4.284250	0.54000000	+1.1233859859e-02
3	114	0.580000	0.290000	4.884250	0.54000000	-1.2327805666e-03
3	115	-0.580000	-0.290000	4.884250	0.54000000	-1.2229792254e-03
3	116	0.580000	-0.290000	4.884250	0.54000000	-1.2662612022e-03
3	117	-0.580000	0.290000	4.884250	0.54000000	-1.2279447029e-03
3	119	-0.000000	-0.000000	5.384250	4.41000000	+4.9499656971e-03

D Verification boundaries and remaining work

Technical claim	Actual status	Appropriate requirement for improvement
824-solid replacement V3 CAD	Conditional physical-design PASS	Measured forming, solder, current and reliability capability
Native pin-to-pin topology	88/88 legs PASS	Finite real current density, resistance and contact temperature fields
Global 430-domain R3.9 mesh	Geometric PASS	Improve worst elements and prove solver conditioning/convergence
Original full-solid electrical Gate B	Historical FAIL/STOP	Qualify original raw-gradient tests on a viable mesh
Full 430-domain electrothermal PDE	NOT RUN	Integrated metal/contact/leg/ceramic field solution
V3.4 Cu-coupled hybrid at 0.4 A	Conditional numerical PASS	Grade/contact uncertainty, spatial metal-sheet fields and current/load response
Rarefied gas and radiation	Effective load model	Native exposed-surface gas/gray/spectral physics
Real thermoelectric grades	Unidentified	Bounded synthetic studies or actual module certificates
HgCdTe photodetection	No carrier/optical PDE	Detector noise/spectral/bias physics if needed
As-built product prediction	Unvalidated	Device measurements; do not claim manufacturer equivalence

Publication policy. The original 38-page Revision 3 remains a sealed separate historical methodology reference. Its later 43-page interim source-mesh integration is likewise archived. This document describes the newer replacement CAD and the most recent completed running hybrid with the unmodeled internal copper paths prominently disclosed. All figures and numerical tables derive from existing saved evidence, and no new physics calculation was performed for this publication.